CORE – 9: Computer Networks (Unit–1)

Introduction to Data Communications and Network Models: A computer network is a set of nodes (devices) connected through links. A node can be computer, printer, or any other device capable of sending or receiving the data. The links connecting the nodes are known as communication channels.

Uses of Computer Network

- **Resource sharing:** Resource sharing is the sharing of resources such as programs, printers, and data among the users on the network without the requirement of the physical location of the resource and user.
- **Client-Server model:** Computer networking is used in the Client-Server model. A server is a central computer used to store the information and maintained by the system administrator. Clients are the machines used to access the information stored in the server remotely.
- **Communication medium:** Computer network behaves as a communication medium among the users. For example, a company contains more than one computer has an email system which the employees use for daily communication.
- **E-commerce:** Computer network is also important in businesses. We can do the business over the internet. For example, amazon.com is doing their business over the internet, i.e., they are doing their business over the internet.

Computer Network Architecture: Computer Network Architecture is defined as the physical and logical design of the software, hardware, protocols, and media of the transmission of data. Simply we can say that how computers are organized and how tasks are allocated to the computer. The two types of network architectures used are:

- Peer-To-Peer network
- Client/Server network

Peer-To-Peer network

- Peer-To-Peer network is a network in which all the computers are linked together with equal privilege and responsibilities for processing the data.
- Peer-To-Peer network is useful for small environments, usually up to 10 computers.
- Peer-To-Peer network has no dedicated server.

Advantages of Peer-To-Peer Network:

- It is less costly as it does not contain any dedicated server.
- If one computer stops working but, other computers will not stop working.
- It is easy to set up and maintain as each computer manages itself.

Disadvantages of Peer-To-Peer Network:

- In the case of Peer-To-Peer network, it does not contain the centralized system. Therefore, it cannot back up the data as the data is different in different locations.
- It has a security issue as the device is managed itself.

Client/Server Network

- Client/Server network is a network model designed for the end users called **clients**, to access the resources from a central computer known as **Server**.
- A server performs all the major operations such as security and network management.
- A server is responsible for managing all the resources such as files, directories, printer, etc.
- All the clients communicate with each other through a server. For example, if client1 wants to send some data to client 2, then it first sends the request to the server for the permission. The server sends the response to the client1 to initiate its communication with the client2.

Advantages of Client/Server network:

- A Client/Server network contains the centralized system. Therefore we can back up the data easily.
- A Client/Server network has a dedicated server that improves the overall performance of the whole system.
- Security is better in Client/Server network as a single server administers the shared resources.
- It also increases the speed of the sharing resources.

Disadvantages of Client/Server network:

- Client/Server network is expensive as it requires the server with large memory.
- A server has a Network Operating System (NOS) to provide the resources to the clients, but the cost of NOS is very high.
- It requires a dedicated network administrator to manage all the resources.

Protocols and Standards: A network **protocol** is an established set of rules that determine how data is transmitted between different devices in the same network. Essentially, it allows connected devices to communicate with each other, regardless of any differences in their internal processes, structure or design.

A **standard** is a generally agreed-upon technology, method or format for a given application. Official standards, also known as **de-jure standards**, are overseen by one of a number of governing bodies that exist to promote their development and confirm their status. The IETF (Internet Engineering Task Force), for example, is an organization that defines Internet operating protocols such as TCP/IP.

De-facto standards, on the other hand, gain prominence because they are widely used. Examples of de facto standards include Microsoft's Windows operating systemt.

Computer Network Models: A communication subsystem is a complex piece of Hardware and software. Early attempts for implementing the software for such subsystems were based on a single, complex, unstructured program with many interacting components. The resultant software was very difficult to test and modify. To overcome such problem, the ISO (International Organization for Standardization) has developed a layered approach. In a layered approach, networking concept is divided into several layers, and each layer is assigned a particular task. Therefore, we can say that networking tasks depend upon the layers.

OSI Model:

- OSI stands for **Open System Interconnection** is a reference model that describes how information from a software application in one computer moves through a physical medium to the software application in another computer.
- OSI consists of seven layers, and each layer performs certain network function.
- OSI model was developed by the International Organization for Standardization (ISO) in 1984, and it is now considered as an architectural model for the inter-computer communications.
- OSI model divides the whole task into seven smaller and manageable tasks. Each layer is assigned specific task.
- Each layer is self-contained, so that task assigned to each layer can be performed independently.

Layers in OSI Model: There are the seven OSI layers. Each layer has different functions. Lists of seven layers are given below:

1. Application layer – Provedes services to the user
2. Presentation layer – Responssible for translation, compression and encryption
3. Session layer – Used to establish, manage and terminate connection
4. Transport layer – Provides reliable messge delivery from process to process
5. Network layer – Responsible for moving the packets from source to destination
6. Datalink layer – Used for error free transfer of data frames
7. Physical layer – Provides a physical medium through bits are trasmitted

Functions of a Physical layer:

- **Line Configuration:** It defines the way how two or more devices can be connected physically.
- **Data Transmission:** It defines the transmission mode whether it is simplex, half-duplex or full-duplex mode between the two devices on the network.
- **Topology:** It defines the way how network devices are arranged.
- **Signals:** It determines the type of the signal used for transmitting the information.

Functions of the Data-link layer

- **Framing:** The data link layer translates the physical's raw bit stream into packets known as Frames. The Data link layer adds the header and trailer to the frame..
- **Physical Addressing:** The header which is added to the frame contains the hardware address of source and destination. The frame is transmitted to the destination address mentioned in the header.
- **Flow Control:** Flow control is the main functionality of the Data-link layer. It is the technique through which the constant data rate is maintained on both the

sides so that no data get corrupted. It ensures that the transmitting station such as a server with higher processing speed does not exceed the receiving station, with lower processing speed.

- **Error Control:** Error control is achieved by adding a calculated value CRC (Cyclic Redundancy Check) that is placed to the Data link layer's trailer which is added to the message frame before it is sent to the physical layer. If any error seems to occurr, then the receiver sends the acknowledgment for the retransmission of the corrupted frames.
- **Access Control:** When two or more devices are connected to the same communication channel, then the data link layer protocols are used to determine which device has control over the link at a given time.

Functions of Network Layer:

- **Internetworking:** An internetworking is the main responsibility of the network layer. It provides a logical connection between different devices.
- **Logical Addressing:** A Network layer adds the source and destination address to the header of the frame. Addressing is used to identify the device on the internet.
- **Routing:** Routing is the major component of the network layer, and it determines the best optimal path out of the multiple paths from source to the destination.
- **Packetizing:** A Network Layer receives the packets from the upper layer and converts them into packets. This process is known as Packetizing. It is achieved by internet protocol (IP).

Functions of Transport Layer:

- **Service-point addressing:** Computers run several programs simultaneously due to this reason, the transmission of data from source to the destination not only from one computer to another computer but also from one process to another process. The transport layer adds the header that contains the address known as a service-point address or port address. The responsibility of the network layer is to transmit the data from one computer to another computer and the responsibility of the transport layer is to transmit the message to the correct process.
- **Segmentation and reassembly:** When the transport layer receives the message from the upper layer, it divides the message into multiple segments, and each segment is assigned with a sequence number that uniquely identifies each segment. When the message has arrived at the destination, then the transport layer reassembles the message based on their sequence numbers.
- **Connection control:** Transport layer provides two services Connection-oriented service and connectionless service. A connectionless service treats each segment as an individual packet, and they all travel in different routes to reach the destination. A connection-oriented service makes a connection with the transport layer at the destination machine before delivering the packets. In connection-oriented service, all the packets travel in the single route.
- **Flow control:** The transport layer also responsible for flow control but it is performed end-to-end rather than across a single link.

- **Error control:** The transport layer is also responsible for Error control. Error control is performed end-to-end rather than across the single link. The sender transport layer ensures that message reach at the destination without any error.

Functions of Session layer:

- **Dialog control:** Session layer acts as a dialog controller that creates a dialog between two processes or we can say that it allows the communication between two processes which can be either half-duplex or full-duplex.
- **Synchronization:** Session layer adds some checkpoints when transmitting the data in a sequence. If some error occurs in the middle of the transmission of data, then the transmission will take place again from the checkpoint. This process is known as Synchronization and recovery.

Functions of Presentation layer:

- **Translation:** The processes in two systems exchange the information in the form of character strings, numbers and so on. Different computers use different encoding methods, the presentation layer handles the interoperability between the different encoding methods. It converts the data from sender-dependent format into a common format and changes the common format into receiver-dependent format at the receiving end.
- **Encryption:** Encryption is needed to maintain privacy. Encryption is a process of converting the sender-transmitted information into another form and sends the resulting message over the network.
- **Compression:** Data compression is a process of compressing the data, i.e., it reduces the number of bits to be transmitted. Data compression is very important in multimedia such as text, audio, video.

Functions of Application layer:

- **File transfer, access, and management (FTAM):** An application layer allows a user to access the files in a remote computer, to retrieve the files from a computer and to manage the files in a remote computer.
- **Mail services:** An application layer provides the facility for email forwarding and storage.
- **Directory services**: An application provides the distributed database sources and is used to provide that global information about various objects.

Analog and Digital Signals: An analog signal is any continuous signal for which the time-varying feature of the signal is a representation of some other time-varying quantity, i.e., *analogous* to another time-varying signal. For example, in an analog audio signal, the instantaneous voltage of the signal varies continuously with the pressure of the sound waves.



Analog Signal



Digital Signal

It differs from a digital signal, in which the continuous quantity is a representation of a sequence of discrete values which can only take on

one of a finite number of values.

Transmission modes: The way in which data is transmitted from one device to another device is known as **transmission mode**. It is also known as the communication mode. The transmission mode is defined in the physical layer.

The Transmission mode is divided into three categories:

- Simplex mode
- Half-duplex mode
- Full-duplex mode

Simplex mode:

- In Simplex mode, the communication is **unidirectional**, i.e., the data flow in one direction.
- A device can only send the data but cannot receive it or it can receive the data but cannot send the data.
- Examples of Simplex mode are **radio, keyboard, and monitor.**
- The main advantage of the simplex mode is that the full capacity of the communication channel can be utilized during transmission.

Half-Duplex mode:

- In a Half-duplex channel, a station can transmit and receive the data as well, but not at the same time.
- The entire bandwidth of the communication channel is utilized in one direction at a time.
- In half-duplex mode, it is possible to perform the error detection, and if any error occurs, then the receiver requests the sender to retransmit the data.
- A **Walkie-talkie** is an example of the Half-duplex mode. In Walkie-talkie, one party speaks, and another party listens. After a pause, the other speaks and first party listens. Speaking simultaneously will create the distorted sound which cannot be understood.
- Here the drawback is that, when one device is sending the data, then another has to wait; this causes the delay in sending the data at the right time.

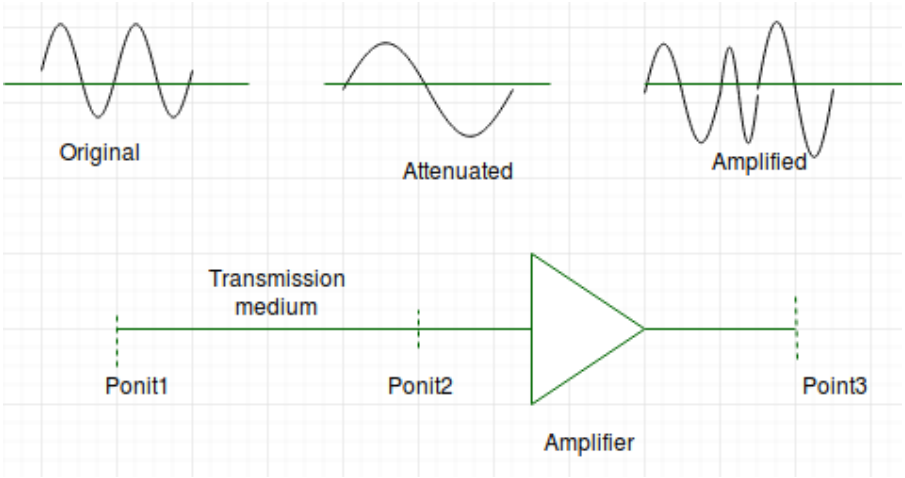
Full-duplex mode

- In Full duplex mode, the communication is bi-directional, i.e., the data flow in both the directions.
- Both the stations can send and receive the message simultaneously.
- The most common example of the full-duplex mode is a telephone network. When two people are communicating with each other by a **telephone line**, both can talk and listen at the same time.

Transmission Impairment: In communication system, analog signals travel through transmission media, which tends to deteriorate the quality of analog signal, which means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. The imperfection causes signal impairment. The **causes of the impairment** are:

- Attenuation
- Distortion
- Noise

Attenuation – It means loss of energy. The strength of signal decreases with increasing distance which causes loss of energy in overcoming resistance of medium. This is also known as attenuated signal.



Amplifiers are used to amplify the attenuated signal which gives the original signal back and compensate for this loss.
Attenuation is measured in decibels (dB). It measures the relative strengths of two signals or one signal at two different points.

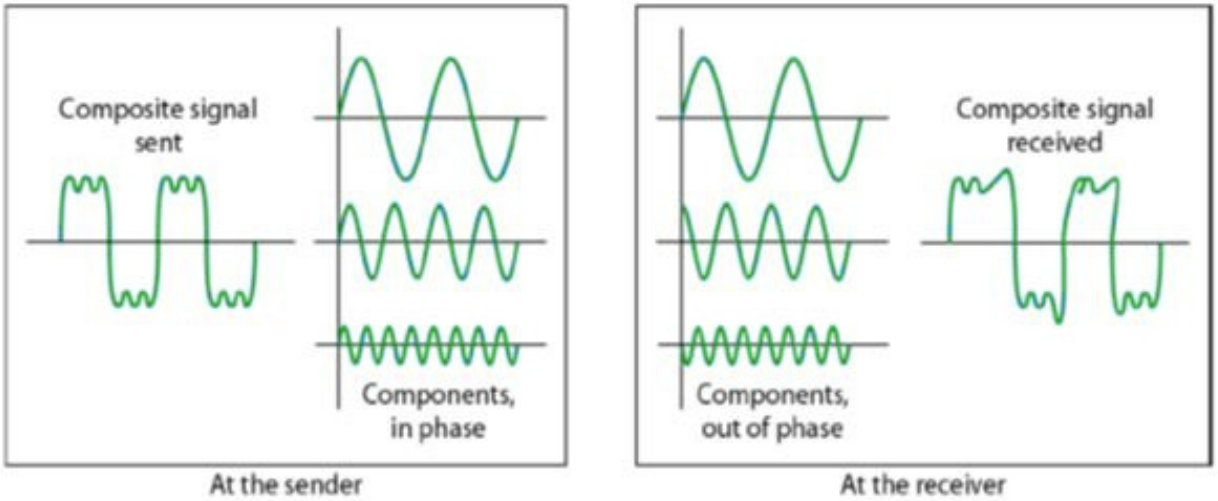
$$\textbf{\textit{Attenuation (dB)} = 10log_{10}(P2/P1)}$$

P1 is the power at sending end and P2 is the power at receiving end.
Some where the decibel is also define in terms of voltage instead of power. In this case, because power is proportional to the square of the voltage the formula is

$$\textbf{\textit{Attenuation(dB)} = 20log_{10}(V2/V1)}$$

V1 is the voltage at sending end and V2 is the voltage at receiving end.

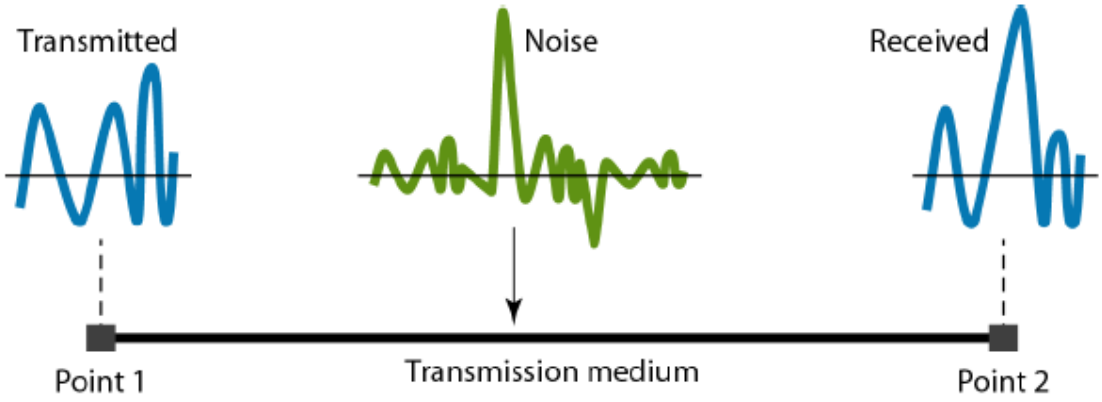
Distortion – It means changes in the form or shape of the signal. This is generally seen in composite signals made up with different frequencies.



Each frequency component has its own propagation speed travelling through a medium and that is why it delays in arriving at the final destination. Every component arrives at different time which leads to distortion. Therefore, they have **different phases** at receiver end from what they had at the sender's end.

Noise – The random or unwanted signal that mixes up with the original signal is called noise. There are several types of noise such as **induced noise, thermal noise, crosstalk noise and impulse noise** which may corrupt the signal. **Induced noise** comes from sources or from the **device itself**. Thermal noise

is movement of electrons in wire which creates an extra signal. **Crosstalk** noise is genereated when **one wire affects the other wire**. **Impulse noise** is a signal with **high energy that comes from lightning** or power lines.



Data Rate Limits: Data rate refers to the speed of data transfer through a channel. It is generally computed in bits per second (bps). Higher data rates are expressed as Kbps ("Kilo" bits per second, i.e.1000 bps), Mbps ("Mega" bits per second, i.e.1000 Kbps), Gbps ("Giga" bits per second, i.e. 1000 Mbps) and Tbps ("Tera" bits per second, i.e. 1000 Gbps). One of the main objectives of data communications is to increase the data rate. There are three factors that determine the data rate of a channel:

- Bandwidth of the channel
- Number of levels of signals that are used
- Noise present in the channel

Data rate can be calculated using two theoretical formulae:

- Nyquist Bit Rate – for noiseless channel
- Shannon's Capacity – for noisy channel

Nyquist Bit Rate: Nyquist bit rate was developed by Henry Nyquist who proved that the transmission capacity of even a perfect channel with no noise has a maximum limit. The theoretical formula for the maximum bit rate is:

$$\textbf{\textit{Maximum bit rate} = 2 \times B \times \log_2 L}$$

Here, *maximum bit rate* is calculated in bps
B is the bandwidth of the channel
L is the number of discrete levels in the signal

For example, if there is a noiseless channel with a bandwidth of 4 KHz that is transmitting a signal with 4 discrete levels, then the maximum bit rate will be computed as, *maximum bit rate* $= 2 \times 4000 \times \log_2 4 = 16{,}000$ *bps = 16 kbps.*

Shannon's Capacity: Shannon extended Nyquist's work for actual channels that are subject to noise. Noise can be of various types like thermal noise, impulse noise, cross-talks etc. Among all the noise types, thermal noise is unavoidable. The random movement of electrons in the channel creates an extraneous signal not present in the original signal, called the thermal noise. The amount of thermal noise is calculated as the ratio of the signal power to noise power, *SNR*.

Signal-to-Noise Ratio (SNR) = Average Signal Power / Average Noise Power

Since *SNR* is the ratio of two powers that varies over a very large range, it is often expressed in decibels, called SNR_{db} and calculated as:

$$\mathbf{SNR_{db} = 10\log_{10}SNR}$$

Shannon's Capacity gives the theoretical **maximum data rate or capacity** of a noisy channel. It is expressed as:

$$\boldsymbol{Capacity\ (C) = B \times \log_2(1+SNR)}$$

Here, *Capacity* is the maximum data rate of the channel in bps
B is the bandwidth of the channel
SNR is the signal-to-noise ratio

For example, if the bandwidth of a noisy channel is 4 KHz, and the signal to noise ratio is 100, then the maximum bit rate can be computed as:

$Capacity = 4000 \times \log_2(1+100) = 26{,}633\ bps = 26.63\ kbps$

Example: We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

Solution: First, we use the Shannon formula to find the upper limit.
$C = B \times \log_2(1+SNR) = 10^6 \log_2(1+63) = 10^6 \log_2 64 = 10^6 \times 6\ bps = 6\ Mbps$
The Shannon formula gives us 6 Mbps, the upper limit. For better performance we choose something lower, 4 Mbps, for example. Then we use the Nyquist formula to find the number of signal levels.

$4\ Mbps = 2 \times 1\ MHz \times \log_2 L$
⇨ $\log_2 L = 2$
⇨ $L = 4$

The Shannon capacity gives us the upper limit; the Nyquist formula tells us how many signal levels we need.

Performance of a Network: Performance of a network pertains to the measure of service quality of a network as perceived by the user. There are different ways to measure the performance of a network, depending upon the nature and design of the network. The characteristics that measure the performance of a network are:

- Bandwidth
- Throughput
- Latency (Delay)
- Bandwidth – Delay Product
- Jitter

Bandwidth: Bandwidth is characterized as the measure of data or information that can be transmitted in a fixed measure of time. The term can be used in two different contexts with two distinctive estimating values. In the case of digital devices, the bandwidth is measured in bits per second (bps). In the case of analog devices, the

bandwidth is measured in cycles per second, or Hertz (Hz). **"Bandwidth" means "Capacity" and "Speed" means "Transfer rate".**

Bandwidth in Hertz*:* It is the range of frequencies contained in a composite signal or the range of frequencies a channel can pass. For example, let us consider the bandwidth of a subscriber telephone line as 4 kHz.

Bandwidth in Bits per Seconds*:* It refers to the number of bits per second that a channel, a link, or rather a network can transmit. For example, we can say the bandwidth of a fast Ethernet network is a maximum of 100 Mbps, which means that the network can send 100 Mbps of data.
Note: There exists an explicit relationship between the bandwidth in hertz and the bandwidth in bits per second. An increase in bandwidth in hertz means an increase in bandwidth in bits per seconds. The relationship depends upon whether we have baseband transmission or transmission with modulation.

Throughput: Throughput is the number of messages successfully transmitted per unit time. It is controlled by available bandwidth, the available signal-to-noise ratio and the hardware limitations. The maximum throughput of a network may be consequently higher than the actual throughput achieved in everyday consumption. The terms 'throughput' and 'bandwidth' are often thought of as the same, yet they are different. **Bandwidth is the potential measurement of a link, whereas throughput is an actual measurement of how fast we can send data**. It is measured in terms of bits transferred per second (bps).

Example: A network with bandwidth of 10 Mbps can pass only an average of 12, 000 frames per minute where each frame carries an average of 10, 000 bits. What will be the throughput for this network?

Solution: Throughput = (12, 000 x 10, 000) / 60 = 2 Mbps
The throughput is nearly equal to one-fifth of the bandwidth in this case.

Latency: The latency or delay defines how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent out from the source. We can say that latency is made of four components: propagation time, transmission time, queuing time and processing delay.

Latency = PropagationTime + TransmissionTime + QueuingTime + ProcessingDelay

Propagation Time: It is the time required for **a bit** to travel from the source to the destination. Propagation time can be calculated as the ratio between the link length (distance) and the propagation speed over the communicating medium.

Propagation time = Distance / Propagation speed

Example: What will be the propagation time when the distance between two points is 12,000 km? Assuming the propagation speed to be 2.4 * 10^8 m/s in cable.

Solution: Propagation time = (12000 * 1000) / (2.4 * 10^8) = 50 ms

Transmission Time: In data communications we don't send just 1 bit, we send a message. The first bit may take a time equal to the propagation time to reach its destination; the last bit also may take the same amount of time. However, there is a

time between the first bit leaving the sender and the last bit arriving at the receiver. The first bit leaves earlier and arrives earlier; the last bit leaves later and arrives later. The time required for transmission of a message depends on the size of the message and the bandwidth of the channel.

Transmission time = Message size / Bandwidth

Example: What will be the propagation time and the transmission time for a 250 kbyte message when the bandwidth of the network is 1 Gbps? Assuming the distance between sender and receiver is 12, 000 km and speed of light is 2.4 * 10^8 m/s.

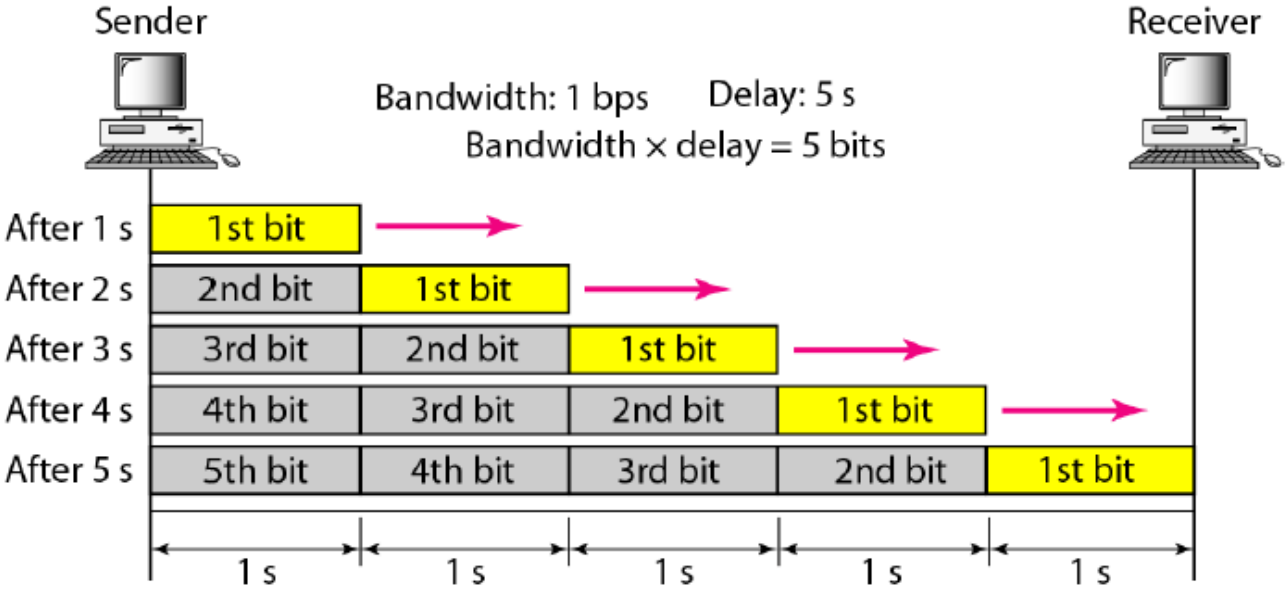
Solution: Propagation time = (12000 * 1000) / (2.4 * 10^8) = 50 ms
Transmission time = (250000 * 8) / 10^9 = 2 ms

Queuing Time: The third component in latency is the queuing time, the time needed for each intermediate or end device to hold the message before it can be processed.

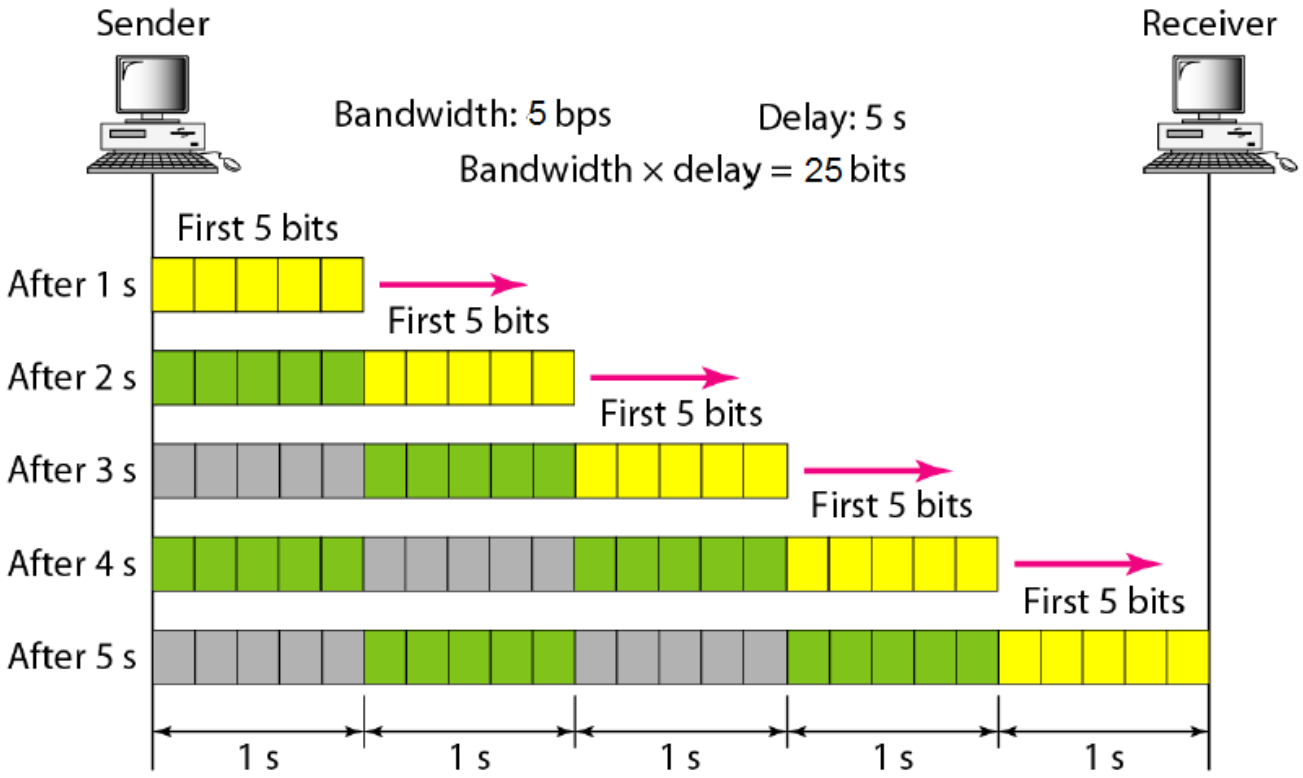
The queuing time is not a fixed factor; it changes with the load imposed on the network. When there is heavy traffic on the network, the queuing time increases. An intermediate device, such as a router, queues the arrived messages and processes them one by one. If there are many messages, each message will have to wait.

Bandwidth Delay Product: Bandwidth and delay are two performance measurements of a link. However, what is significant in data communications is the product of the two, the bandwidth-delay product. Let us take two hypothetical cases as examples.

Case 1: Assume a link is of bandwidth 1bps and the delay of the link is 5s. Let us find the bandwidth-delay product in this case. From the image, we can say that this product 1 x 5 is the maximum number of bits that can fill the link. There can be close to 5 bits at any time on the link.



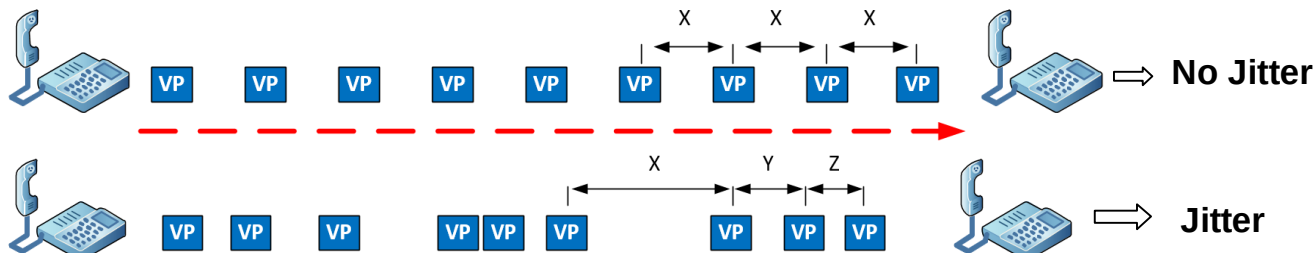
Case 2: Assume a link is of bandwidth 5bps. From the image, we can say that there can be a maximum 5 x 5 = 25 bits on the line. The reason is that, at each second, there are 5 bits on the line and the duration of each bit is 0.2s.



The above two cases show that the product of bandwidth and delay is the number of bits that can fill the link. This measurement is important if we need to send data in bursts and wait for the acknowledgment of each burst before sending the next one. To use the maximum capability of the link, we need to make the size of our burst twice the product of bandwidth and delay; we need to fill up the full-duplex channel (two directions). The sender should send a burst of data of (2 x bandwidth x delay) bits. The sender then waits for receiver acknowledgment for part of the burst before sending another burst. The amount 2 x bandwidth x delay is the number of bits that can be in transition at any time.

The bandwidth-delay product defines the number of bits that can fill the link.

Jitter: Jitter is another performance issue related to delay. In technical terms, jitter is a "**packet delay variance**". It can simply mean that jitter is considered as a problem when different packets of data face different delays in a network and the data at the receiver application is time-sensitive, i.e. audio or video data. Jitter is measured in milliseconds (ms). It is defined as interference in the normal order of sending data packets. For example: if the delay for the first packet is 10 ms, for the second is 35 ms, and for the third is 50 ms, then the real-time destination application that uses the packets experiences jitter. **Jitter is negative and causes network congestion and packet loss.**



Jitter occurs when voice packets (labeled vp in the diagram) do not arrive with consistent periodicity. A **jitter buffer can reduce** the effects of jitter, either in a network, on a router or switch, or on a computer. The system at the destination receiving the network packets usually receives them from the buffer and not from the source system directly. Each packet is fed out of the buffer at a regular rate. Another approach to diminish jitter in case of multiple paths for traffic is to **selectively route traffic** along the most stable paths, or to always pick the path that can come closest to the targeted packet delivery rate.

Digital Transmission: Data can be represented either in analog or digital form. The computers used the digital form to store the information. Therefore, the data needs to be converted in digital form so that it can be used by a computer.

Network Devices & Drivers:

1. **Repeater** – A repeater operates at the **physical layer**. Its job is to regenerate the signal over the same network before the signal becomes too weak or corrupted so as to extend the length to which the signal can be transmitted over the same network. An important point to be noted about repeaters is that they do not amplify the signal. When the signal becomes weak, they copy the signal bit by bit and regenerate it at the original strength. It is a 2 port device.
2. **Hub** – A hub is basically a multiport repeater. A hub connects multiple wires coming from different branches, for example, the connector in star topology which connects different stations. Hubs cannot filter data, so data packets are sent to all connected devices. Also, they do not have the intelligence to find out best path for data packets which leads to inefficiencies and wastage.

Types of Hub

- **Active Hub: -** These are the hubs which can clean, boost, and relay the signal along with the network. These are used to extend the maximum distance between nodes.
- **Passive Hub: -** These hubs relay signals onto the network without cleaning and boosting them and can't be used to extend the distance between nodes.
- **Intelligent Hub: -** It work like active hubs and include remote management capabilities. It also enables an administrator to monitor the traffic passing through the hub and to configure each port in the hub.

3. **Bridge** – A bridge operates at **data link layer**. A bridge is a repeater; with add on the functionality of filtering content by reading the MAC addresses of source and destination. It is also used for interconnecting two LANs working on the same protocol. It has a single input and single output port, thus making it a 2 port device.

Types of Bridges

- **Transparent Bridges: -** These are the bridge in which the stations are completely unaware of the bridge's existence i.e. whether or not a bridge is added or deleted from the network, reconfiguration of the stations is unnecessary. These bridges make use of two processes i.e. bridge forwarding and bridge learning.

- **Source Routing Bridges: -** In these bridges, routing operation is performed by source station and the frame specifies which route to follow. The host can discover frame by sending a special frame called discovery frame, which spreads through the entire network using all possible paths to destination.

4. **Switch** – A switch is a multiport bridge with a buffer and a design that can boost its efficiency (a large number of ports imply less traffic) and performance. A switch is a **data link layer device**. The switch can perform error checking before forwarding data that makes it very efficient as it does not forward packets that have errors and forward error-free packets selectively to correct port only. In other words, switch divides collision domain of hosts, but broadcast domain remains same.

5. **Routers** – A router is a device like a switch that routes data packets based on their IP addresses. Router is mainly a **Network Layer device**. Routers normally connect LANs and WANs together and have a dynamically updating routing table based on which they make decisions on routing the data packets. Router divide broadcast domains of hosts connected through it.

6. **Modem** – A modem is a hardware device that allows the computer to connect to the internet over the existing telephone line. A modem is not integrated with the motherboard rather than it is installed on the PCI slot found on the motherboard. It stands for Modulator/Demodulator. It converts the digital data into an analog signal over the telephone lines. Based on the differences in speed and transmission rate, a modem can be classified into the following categories:

    - Standard PC modem or Dial-up modem
    - Cellular Modem
    - Cable modem